

CLOUD COMPUTING:

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EX NO:38 to 40 Hadoop installation and programs:

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cloud@cloud-VMware-Virtual-Platform: ~  
  
cloud@cloud-VMware-Virtual-Platform:~$ start-dfs.sh  
Starting namenodes on [localhost]  
Starting datanodes  
Starting secondary namenodes [cloud-VMware-Virtual-Platform]  
cloud@cloud-VMware-Virtual-Platform:~$ start-yarn.sh  
Starting resourcemanager  
Starting nodemanagers  
  
at org.apache.hadoop.util.RunJar.run(RunJar.java:326)  
at org.apache.hadoop.util.RunJar.main(RunJar.java:254)  
cloud@cloud-VMware-Virtual-Platform:~$ hadoop jar wordcount.jar WordCount /wordcount/input /wordcount/output12  
2026-08-26 10:31:38,046 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032  
2026-08-26 10:31:39,289 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.  
2026-08-26 10:31:39,334 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/cloud/.staging/job_1787720296530_0001  
2026-08-26 10:31:40,440 INFO input.FileInputFormat: Total input files to process : 1  
2026-08-26 10:31:40,944 INFO mapreduce.JobSubmitter: number of splits:1  
2026-08-26 10:31:41,365 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1787720296530_0001  
2026-08-26 10:31:41,365 INFO mapreduce.JobSubmitter: Executing with tokens: []  
2026-08-26 10:31:41,665 INFO conf.Configuration: resource-types.xml not found  
2026-08-26 10:31:41,666 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.  
2026-08-26 10:31:42,544 INFO impl.YarnClientImpl: Submitted application application_1787720296530_0001  
2026-08-26 10:31:42,743 INFO mapreduce.Job: The url to track the job: http://cloud-VMware-Virtual-Platform:8088/proxy/application_1787720296530_0001/  
2026-08-26 10:31:42,751 INFO mapreduce.Job: Running job: job_1787720296530_0001  
2026-08-26 10:31:55,131 INFO mapreduce.Job: Job job_1787720296530_0001 running in uber mode : false  
2026-08-26 10:31:55,136 INFO mapreduce.Job: map 0% reduce 0%  
2026-08-26 10:32:02,304 INFO mapreduce.Job: map 100% reduce 0%  
2026-08-26 10:32:09,432 INFO mapreduce.Job: map 100% reduce 100%  
2026-08-26 10:32:10,471 INFO mapreduce.Job: Job job_1787720296530_0001 completed successfully  
2026-08-26 10:32:10,600 INFO mapreduce.Job: Counters: 54  
File System Counters  
FILE: Number of bytes read=70  
FILE: Number of bytes written=634471
```

```
cloud@cloud-VMware-Virtual-Platform: ~  
FILE: Number of read operations=0  
FILE: Number of large read operations=0  
FILE: Number of write operations=0  
HDFS: Number of bytes read=146  
HDFS: Number of bytes written=27  
HDFS: Number of read operations=8  
HDFS: Number of large read operations=0  
HDFS: Number of write operations=2  
HDFS: Number of bytes read erasure-coded=0  
Job Counters  
  Launched map tasks=1  
  Launched reduce tasks=1  
  Data-local map tasks=1  
  Total time spent by all maps in occupied slots (ms)=5075  
  Total time spent by all reduces in occupied slots (ms)=4513  
  Total time spent by all map tasks (ms)=5075  
  Total time spent by all reduce tasks (ms)=4513  
  Total vcore-milliseconds taken by all map tasks=5075  
  Total vcore-milliseconds taken by all reduce tasks=4513  
  Total megabyte-milliseconds taken by all map tasks=5196800  
  Total megabyte-milliseconds taken by all reduce tasks=4621312  
Map-Reduce Framework  
  Map input records=1  
  Map output records=5  
  Map output bytes=54  
  Map output materialized bytes=70  
  Input split bytes=112  
  Combine input records=0
```

```
cloud@cloud-VMware-Virtual-Platform: ~  
  Reduce input records=3  
  Reduce output records=3  
  Spilled Records=10  
  Shuffled Maps =1  
  Failed Shuffles=0  
  Merged Map outputs=1  
  GC time elapsed (ms)=61  
  CPU time spent (ms)=1920  
  Physical memory (bytes) snapshot=463470592  
  Virtual memory (bytes) snapshot=5505490944  
  Total committed heap usage (bytes)=255852544  
  Peak Map Physical memory (bytes)=276094976  
  Peak Map Virtual memory (bytes)=2750189568  
  Peak Reduce Physical memory (bytes)=187375616  
  Peak Reduce Virtual memory (bytes)=2755301376  
Shuffle Errors  
  BAD_ID=0  
  CONNECTION=0  
  IO_ERROR=0  
  WRONG_LENGTH=0  
  WRONG_MAP=0  
  WRONG_REDUCE=0  
File Input Format Counters  
  Bytes Read=34  
File Output Format Counters  
  Bytes Written=27  
cloud@cloud-VMware-Virtual-Platform:~$ hdfs dfs -cat/wordcount/output12/part-r-00000  
-cat/wordcount/output12/part-r-00000: Unknown command  
Usage: hadoop fs [generic options]
```

-libjars <jar1,...> specify a comma-separated list of jar files to be included in the classpath
-archives <archive1,...> specify a comma-separated list of archives to be unarchived on the compute machines

The general command line syntax is:
command [genericOptions] [commandOptions]

```
cloud@cloud-VMware-Virtual-Platform:~$ hdfs dfs -cat /wordcount/output12/part-r-00000
```

```
bigdata 1
hadoop 2
hello 2
```

```
cloud@cloud-VMware-Virtual-Platform:~$ ls
```

Desktop	Pictures	'WordCount\$IntSumReducer.class'	a.out	input.txt	sample.txt
Documents	Public	'WordCount\$TokenizerMapper.class'	file1	project	snap
Downloads	Templates	WordCount.class	file1welcome	sam.c	wordcount.jar
Music	Videos	WordCount.java	hadoop-3.5.0.tar.gz	sample.c	

```
cloud@cloud-VMware-Virtual-Platform:~$ cat input.txt
```

```
hello hadoop hello bigdata hadoop
```

```
cloud@cloud-VMware-Virtual-Platform:~$
```